

S1 G8 ITS Milestone-04 Scribe

School of Engineering and Applied Science (SEAS)
Ahmedabad University

CSE 400: Fundamentals of Probability in Computing

Group: Section-01, Group-08, ITS

Group Members:

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Scribe Question 1: Baseline System and Motivation for Randomization

Briefly restate the deterministic baseline implemented in Milestone-3. Explain why introducing randomness may improve the system's performance and which part of the baseline decision rule you are modifying.

In the previous Milestone (Milestone-03), our group implemented one variant of the Dijkstra algorithm which is a Deterministic type of algorithm for the baseline analysis of our given problem SOTA (Stochastic On Time Arrival).

Now, the main limitation of a deterministic baseline arrives with the name of our problem where Stochastic means Random. As a result, we can understand that what we are dealing with does not revolve around deterministic ironclad decisions only, we are also considering the part where the probabilistic analysis is involved. Which basically means that using randomized algorithm is the only thing which actually lifts the result of the deterministic algorithm and lets us observe it from another perspective. That perspective enables us to ignore the shortest path based on the distance and lets us choose the shortest path based on the reliability of arriving under a certain fixed time budget.

Scribe Question 2: Randomized Algorithm Design

Describe the randomized algorithm you have designed.

- What decision component now involves randomness?
- What probability distribution or sampling mechanism is used?
- How does this randomized rule differ from the deterministic baseline?

Provide a clear explanation of the algorithm.

The randomized algorithm designed for the Stochastic On Time Arrival (SOTA), is based on Time-Budget Routing instead of taking fixed value of time for traveling a particular edge it considers the probability of crossing the vertices according to the probability of the given road edge. In easier terms it is purely reflecting the dependency or the reliability of a particular path rather than giving us the fixed time like Deterministic Algorithm. That is where the randomness arrives from. The random component of the whole analysis is nothing but the PMF of different edges.

Normally, the ideal sampling mechanism for our simulation would be the Gaussian Distribution. Which is already incorporated into the simulation but just to see the variation in the results, we have also incorporated different sampling techniques with respect to the results of the Randomized Algorithm. Which includes Bimodal Sampling for Rush Hour Scenario, Uniform for Random Sampling as well as a Long-tail Sampling for a semi predictable path curve.

Our approach differs from the deterministic baseline in many ways but the most important part is that we are focusing on a very compact yet subtle heuristic approach. Which means that we are also considering the factors of predictability or uncertainty in our randomized analysis which would have been not considered normally in binary decision of making of deterministic result of Milestone-03.

Scribe Question 3: Probabilistic Reasoning Behind the Algorithm

Explain the probabilistic reasoning that motivates your randomized approach.

- How does randomness help address uncertainty in the system?
- What probabilistic assumptions or models support your design?
- Why is this randomized strategy expected to improve performance?

The main probabilistic principle used in our randomized approach is choosing the most reliable path over the most shortest path because in real world applications the shortest path is not always the most beneficiary. Hence, the randomized approach may perform slower than the deterministic approach to give the answer but the path it gives would be more reliable or optimum compared to the deterministic algorithm.

Randomness is found on the edges of 2 or more vertices, where the reliability and the travel time are all different for every path. Now, when our algorithm starts comparing the components, it actually assumes all the different edges or road networks as individual probability mass distributions, hence it actually compares there PMF directly to get the single most reliable or optimum path out of all the others.

Normally, the deterministic algorithm would look faster compared to the randomized approach but the main factor of comparison between these algorithms is not their respective time complexities only, it is the quality of the answers they provide. The randomized does not only consider the factors which are almost inconsiderable for the deterministic algorithm but it also compares the multiple graph networks with each other while assuming them all to be independent probability distributions by the method of convolution simultaneously.

Scribe Question 4: Implementation within the Simulation Pipeline

Describe how the randomized algorithm has been integrated into your existing simulation pipeline.

- What changes were made to the code or system architecture?
- How is randomness generated or sampled in the implementation?
- Does the rest of the pipeline remain the same as the baseline?

The code for the randomized algorithm was already provided by the authors of our reference paper but the code which they provided was a little different in input, output and processing technique. As a result, we did some changes in the code to make it more adaptable to the mini artificial datasets generated by us for testing purposes.

The code was rewritten with the same algorithm but the dataset processing was a different than before in Milestone-02. That code was now updated for the Milestone-04 for integrating it in a website for visualization of the comparison of the deterministic and the randomized approach.

The randomness is generated by a python script for the graph generation process. Then it is fed to the deterministic and the randomized approaches simultaneously in parallel pipelining. A time budget is also allocated which is same for the both the algorithms with a source and destination ID for the algorithms to work on. Then when started we can actually observe the live time path chosen by the algorithms for comparison with graph and probability of arriving under the time limit if we choose the path given by both the algorithms separately for further comparison.

Partially, the whole pipeline is similar to the baseline but it can change depending on the use of algorithms or the input variables. We can also use different sampling method to differentiate the current pipeline from the baseline which only considers the Gaussian Probability Distribution only. Overall around half of the pipeline resembles that of the baseline.

Scribe Question 5: Evaluation Metrics and Comparative Results

Identify the evaluation metrics used to measure system performance.

- Why are these metrics appropriate for your problem domain?
- How does the randomized algorithm perform compared to the deterministic baseline?
- Present and interpret the results of this comparison.

The evaluation metrics for our system pipeline would be the careful analysis of the chosen path network in the graph as well as the comparison of the probability of arriving from the source to the destination and the allocated budget time. These two main factors describe the quality of the overall result derived from the analysis of the randomization process of the algorithm.

The randomized algorithm as stated in the above question is a little slower than the deterministic baseline but the most significant difference can be observed as the resultant path given to us by the randomized algorithm. The deterministic approach, specifically LET (Least Expected Time) version of Dijkstra is a greedy type of algorithm which is nothing but a subset of divide and conquer technique. While the randomized approach is also based on comparison between the paths but not their distance, the comparison happens only when the particular vertex or the edge is taken as an independent probability distribution function itself based on the reliability complex of the suggested path. These both algorithms are very different in reality even if they seem similar on the surface.

The result of a simulation of a graph with 100 vertex and a budget time of 550 seconds can be seen below in the image form. The left side is the Dijkstra (Deterministic) and the right side of the screen is the A* Heuristic (Randomized) algorithm visualizers respectively. The test was done on the website created for visualization and comparison for the algorithms.

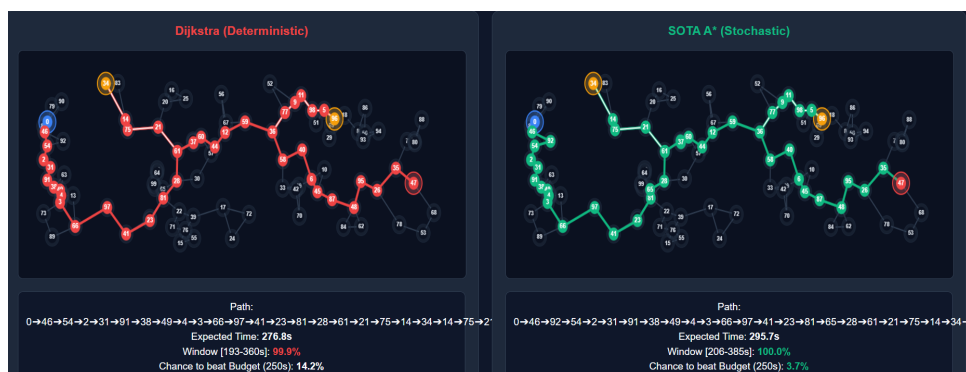


Figure 1: Credits: https://codesensei404.github.io/G8_S1_ITS/

Scribe Question 6: Insights and Remaining Challenges

Based on your experiments:

- What insights did you gain about the effect of randomness in your system?
- In what situations does the randomized strategy perform better or worse than the baseline?

In our system, the effects of randomness taught us that how randomized algorithm have the unparalleled efficiency in terms of the quality of the answer for our case of Intelligence Transportation System. The deterministic approach is without doubt faster than our randomized approach but it is pure comparison on the shortest path, while the Randomized approach considers the uncertainty itself.

In a smaller graph network (≤ 100 Nodes), both of the algorithm almost gives solution with the same time. But for larger input size the time complexity of the randomized algorithm increases faster compared to the Dijkstra. Though, the time taken by the randomized algorithm is greater than that of the deterministic approach but the primary factor here to consider is the probability of arriving within the time budget or the reliability of the chosen path network which is fulfilled by the randomized algorithm. Hence, in terms of a good and fast solution we can use the deterministic approach but for it the answer to be great and realistic we must only use the randomized approach.